The AI/ML Interview Drop

ISSUE #1 · NOV 12, 2026 · 15 QUESTIONS

Linear Regression, L1 vs L2 and Transformer Tensors

How to use this: answer each question out loud before reading its hint, then tick the box if you could explain it clearly in two minutes.

ML Fundamentals · 3 | Deep Learning · 1 | LLMs & Transformers · 3 | RAG & Agents · 3 | MLOps & Production · 1 | ML / AI System Design · 2 | Stats & Math · 1 | Coding & Implementation · 1

DEEP DIVE

Walk me through what happens, tensor by tensor, in one forward pass of a decoder-only transformer.

We begin with an input token sequence of shape (batch_size, sequence_length), which passes through an embedding layer to yield tensor X of shape (batch_size, seq_len, d_model).

Next, X enters the transformer block layers. Each layer applies RMSNorm or LayerNorm, followed by self-attention projections. We compute Query, Key, and Value tensors by multiplying X with weight matrices, reshaping to multi-head format (batch, heads, seq_len, d_head), applying causal masking, and computing scaled dot-product attention.

The attention output tensor is projected back to d_model, added to the original X via a residual connection, and normalized again. This tensor then passes through the feed-forward network, typically utilizing gated activations like SwiGLU, expanding hidden dimensions before projecting back to d_model.

After passing through all stacked decoder layers, the final tensor is normalized and projected via the language model head weight matrix to produce logits of shape (batch_size, seq_len, vocab_size), predicting the next token probabilities.

ML FUNDAMENTALS

01 What is the closed-form solution of linear regression, and when do you use gradient descent instead? ☐

Easy · via GitHub · ai engineering interview questions company wise

A strong answer covers: Mention the normal equation and its cubic time complexity with respect to features. Contrast this with gradient descent for massive datasets.

02 What are the differences between L1 and L2 regularization in logistic regression? ☐

Easy · via GitHub · ai engineering interview questions company wise

A strong answer covers: Explain sparsity via absolute penalties versus shrinkage via squared penalties, and how they behave with correlated features.

03 A network confuses pugs and pit bulls and some training labels are wrong. How do you modify the model and the data?

Medium · **Asked this week** · via GitHub · ai engineering interview questions company wise

A strong answer covers: Address noisy labels using robust loss functions or confidence weighting, and handle fine-grained visual classes using attention maps or metric learning.

DEEP LEARNING

04 You need one embedding space for your own domain: product photos, spec sheets as PDFs, and text queries. Off-the-shelf CLIP is not good enough. How do you build it?

Hard · via GitHub · AI Engineer Interview Questions

A strong answer covers: Discuss multi-modal joint embedding spaces, PDF parsing strategies, contrastive fine-tuning, and hard negative mining.

LLMS & TRANSFORMERS

05 GLM-4.5 is an MoE with 355B total but 32B active parameters. Explain the economics: what does that split buy you and what does it cost?

Medium · via GitHub · ai engineering interview questions company wise

A strong answer covers: Analyze inference capacity scaling, memory bandwidth bottlenecks, routing overhead, and pre-training throughput tradeoffs.

06 Walk me through what happens, tensor by tensor, in one forward pass of a decoder-only transformer.

Hard · **Asked this week** · via X

A strong answer covers: Trace tokenization, embedding lookup, multi-head attention projections, residual additions, layer normalization, and feed-forward transformations.

07 Your DPO run improved general chat evaluations but regressed math performance. What are your hypotheses for why this happened?

Hard · **Asked this week** · via github.com

A strong answer covers: Explore reward hacking, catastrophic forgetting, and distribution shifts between conversational style data and precise reasoning benchmarks.

RAG & AGENTS

08 How would you benchmark an LLM agent's tool use, say, for enterprise workflows composing 10+ APIs?

Medium · **Asked this week** · via GitHub · ai engineering interview questions company wise

A strong answer covers: Propose multi-turn execution environments, deterministic mock APIs, correctness verification, and cost-latency profiling frameworks.

09 **Users ask “how many open orders are blocked on a supplier issue?” Plain RAG gets this wrong. Why, and what's the right architecture?**

Medium · via GitHub · ai engineering interview questions company wise

A strong answer covers: Highlight aggregation limitations of unstructured search, and advocate for Text-to-SQL or hybrid structured-unstructured retrieval pipelines.

10 **The patient's chart already lists their medications. How would you use that to improve transcription of drug names, and how would you keep it from backfiring?**

Medium · via GitHub · ai engineering interview questions company wise

A strong answer covers: Cover biasing speech recognition decoding graphs with patient vocabularies, alongside mitigation for hallucinations when charts are incomplete.

MLOPS & PRODUCTION

11 **How do you load test an LLM service so the numbers actually mean something?**

Hard · via GitHub · AI Engineer Interview Questions

A strong answer covers: Focus on realistic workload traffic patterns, TTFT versus ITL metrics, concurrency ramps, and avoiding synthetic client-side bottlenecks.

ML / AI SYSTEM DESIGN

12 **When should you choose fine-tuning over RAG over prompt engineering?**

Medium · via GitHub · ai engineering interview questions

A strong answer covers: Weigh cost, latency, update frequency, domain knowledge requirements, and security constraints across architectural choices.

13 **Why might replacing a dense feed-forward layer with a top-2 Mixture of Experts (MoE) increase end-to-end inference latency despite executing fewer FLOPs per token?**

Hard · **Asked this week** · via X

A strong answer covers: Mention memory bandwidth limits, non-contiguous expert memory accesses, dynamic batching fragmentation, and all-to-all communication overhead.

STATS & MATH

14 **What are Type I and Type II errors?**

Easy · via GitHub · machine learning interview questions

A strong answer covers: Define false positives and false negatives clearly in the context of hypothesis testing or classification thresholds.

CODING & IMPLEMENTATION

15 **Implement beam search for an autoregressive model. When would an answer engine actually use it?**

Medium · via GitHub · ai engineering interview questions company wise

A strong answer covers: Describe probability tracking, pruning, and length penalties. Explain its use in high-stakes structured generation tasks.

Practise this issue online, with hints hidden and your progress saved: shubham-00787.github.io/ai-interview-drop/issues/1.html